

BLUEPRINT READING FOR WELD 65



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Blueprint Reading

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Licensing

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CHAPTER OVERVIEW

Modules

- 1: Visualization
- 2: Auxiliary Views
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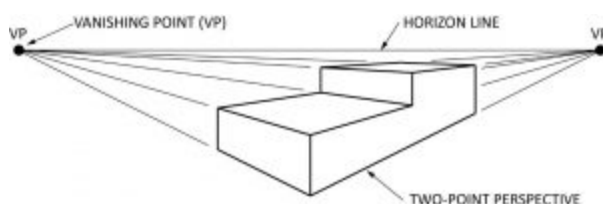
1: Visualization

Now that you have learned about the kinds of lines found on prints, the next step is to develop your visualization abilities. The ability to "see" technical drawings; that is to "think in three dimensions," is the most important part of this course. Since most engineering and architectural prints utilize some form of orthographic projection (multi-view drawing), that type of drawing will be emphasized.

Before going into a study of orthographic projection, you should be able to recognize several other types of drawings. They are; 1. Perspective drawing, 2. Oblique drawing, and 3. Isometric drawing. As a group, they are called "pictorial drawings". They are found on prints and are easy to visualize, so let's look at their differences.

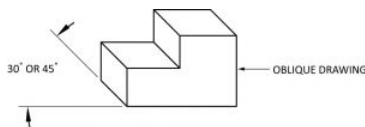
Perspective

Perspective is the most realistic form of drawing. Artists use one-point perspective, two-point (shown here), and three point to create visual depth. Perspectives are used by architects and for industrial pictorials of plan layouts, machinery, and other subjects where realism is required. Objects drawn in perspective grow smaller as they recede into the horizon.



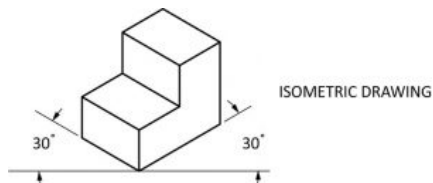
Oblique

Oblique drawings are drawn with one plane (front) of the object parallel to the drawing surface. The side, or other visible part of the object is generally drawn at 30° or 45° . Note that only the side is on an angle. Many times these types of drawing are not drawn to scale. The receding lines are drawn at 45° or 30° and will be drawn at a different scale as the vertical and horizontal lines. This make the drawing seem "out of shape". This type of drawing is not used very often in industry.

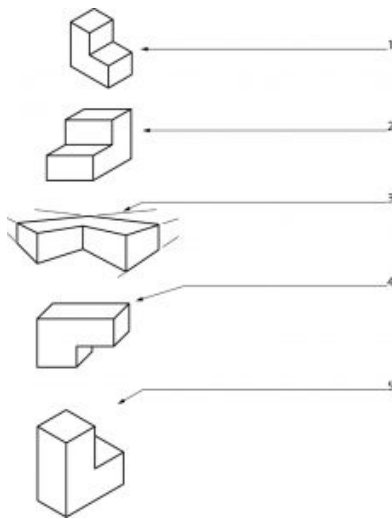


Isometric

Isometric drawings have less distortion than oblique drawings, and are used more frequently by industry for that reason. An isometric drawing has both visible surfaces drawn at 30° . These are the most used type of drawings in the piping industry and take a good deal of practice to fully understand how to draw. They best represent what is being built and what it will look like from the different sides with one drawing.



Directions: Name the types of drawings shown below. Check your own answers.

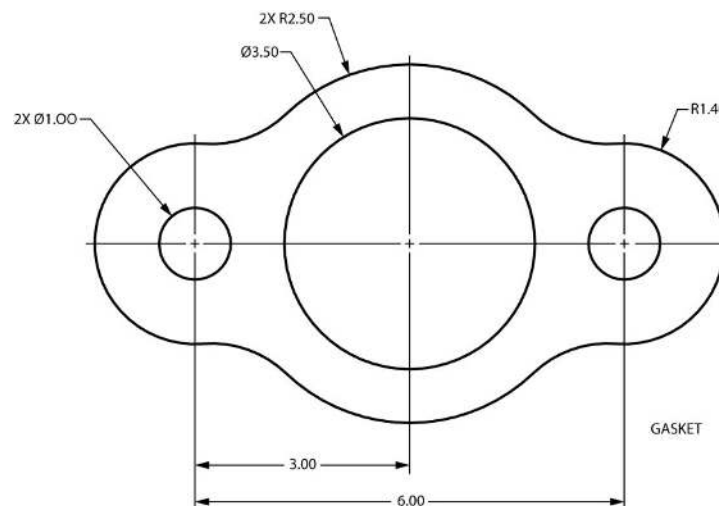


Single View

A single view of an object is sometimes all that is needed for a complete visual explanation. When dimensions, material, and other information is Included, an object requiring only a single view is easy to understand.

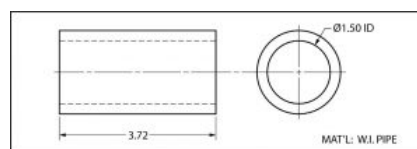
Most one-view drawings are of flat objects, made from materials such as sheet metal and gasket stock. Spherical objects, such as a cannonball, would require only one view and a note indicating the material and diameter of the sphere.

The object shown in the one-view drawing below could be made of any appropriate material that might be specified. In appearance, it is much like the gasket used as part of the cooling system on many cars. All that would need to be noted is the material type and thickness required.



Two View

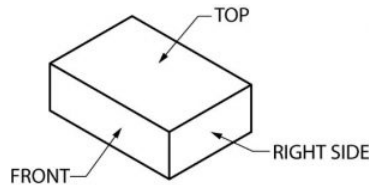
Sometimes “two-view” drawings are use on prints. Two views may be all that is needed to show the shape of an object. Objects that are cylindrical, such as a length of pipe, are usually shown on a print with two views. In such a case, two views is sufficient to explain the shape. Notice in the two-view drawing shown below that the length of the pipe is shown in one view, while the diameter is called out in the other. Without the view on the right, what might this shape be mistaken for? Square tube, channel...



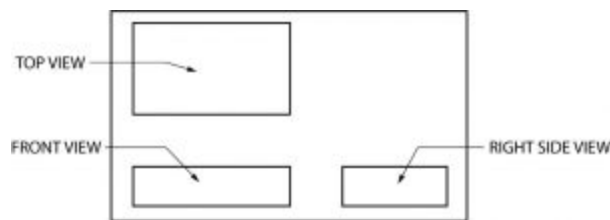
Orthographic Projection

Orthographic projection is a name given to drawings that usually have three views. Often, the three views selected are the top, front, and right side. It is possible, of course, to select other views such as the left side or bottom. Generally, though, it's the top, front and right side that are traditionally seen by the person reading prints.

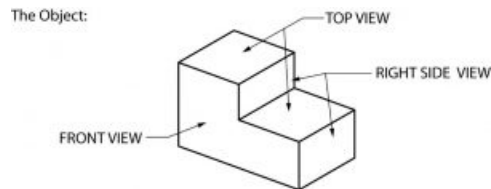
Since most prints make use of the orthographic projection system, and because the top, front, right side views are most often used, it is important that you have their order, or arrangement on the print fixed in your mind. To help you understand this system, think of a chalkboard eraser, a short length of 2" x 4" lumber, or a common brick. It looks like this:



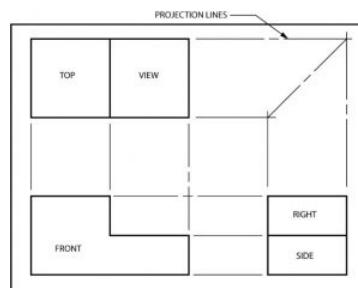
When seen on a print, using orthographic projection, it would appear like this.



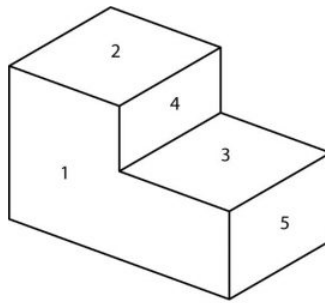
This system of orthographic projection may be difficult to understand or visualize at first, but you will grasp it with some practice. Here's a basic example of how it works, using a simple object.



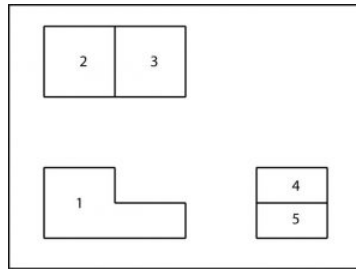
Orthographic projection does not show depth, so the object shown above will appear flat. With practice, however, you will learn to scan the three views and "read" depth into them. Remember that the location of the top, front and right side views does not change. The projection lines between the orthographic views below show the height, width, and depth relationship that exists between each view and the other two views.



In case you did not understand the three-view on the last page, let's take another look at the same thing. This time numbers will be used for identification for the surfaces.



Using orthographic projection, the object with the surfaces numbered appears like this:



Notice that the front view (1) is the key to the drawing, because it most clearly shows the shape of the object. It tells you the object is “L” shaped from the front. The other two views don’t tell you much by themselves. By looking at surface 1, however, you can see that 2 is taller than 3. Therefore, in “reading” the surfaces, 2 should appear to be closer to you than 3. Now look at 4 and 5. Which surface is projected closest to you?

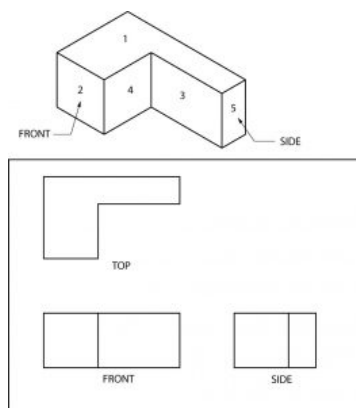
Answer: Surface 5 (rotate and place at bottom of layout)

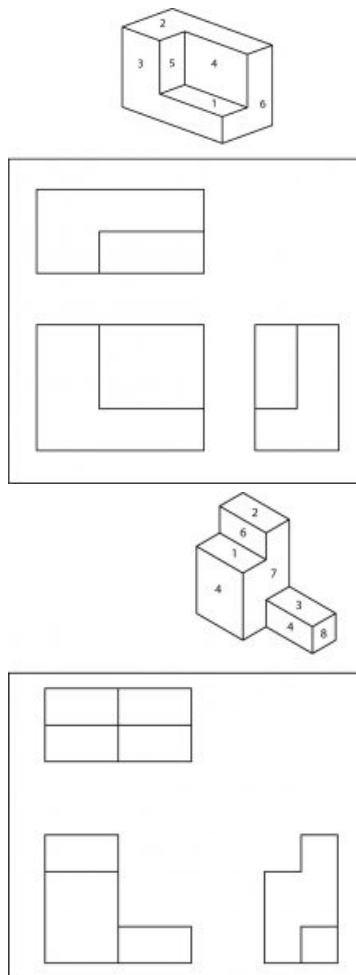
Now draw a simple box and tape all sides together to form a cube. The cube will be 2”x 2”x 2”. Once the instructor has approved your drawing you will proceed to cut out and tape edges together to form a cube.

Visualization Quiz

Directions:

All visible surfaces on the objects shown are numbered. To complete this quiz, you are to place those numbers on the corresponding surfaces of the orthographic drawings.





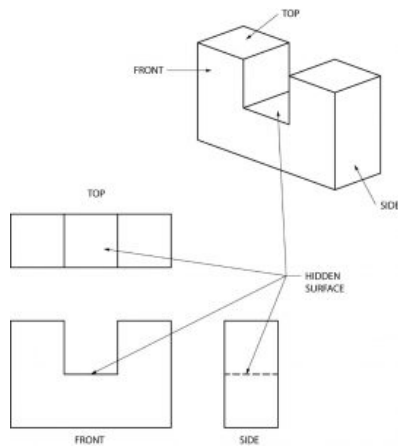
You may be wondering at this point why something like orthographic projection is used on prints when isometric or oblique drawings are so much easier to visualize. The answer is that both of those types of pictorials are used for relatively uncomplicated drawings. When an object is complex, however, neither can equal the orthographic system for clear presentation of dimensions, notes, and configuration details.

Hidden Surfaces

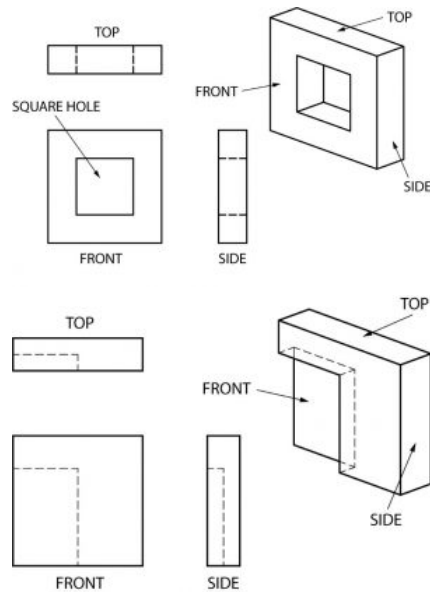
Another advantage of orthographic projection is that it allows the person reading the print to have the ability to see the inside, or surfaces of an object which normally could not be seen.

With complicated objects this can become very useful.

In the drawing below, the hidden line in the right side view represents the entire surface of the flat area between the two higher sides.



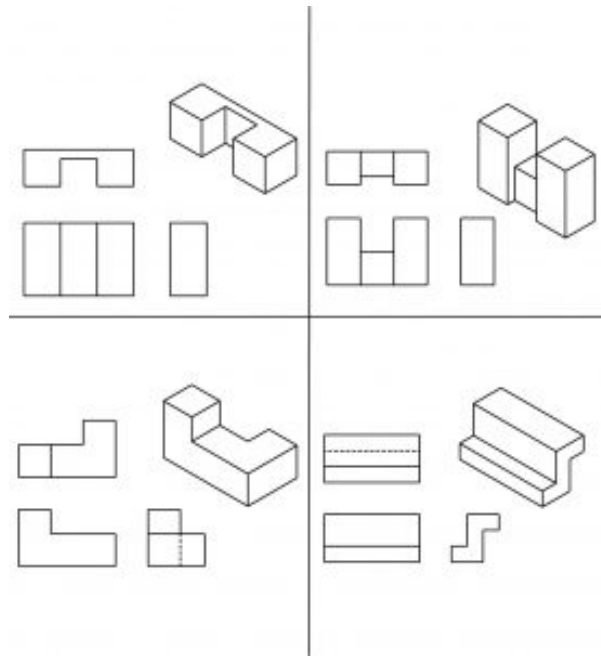
In this example, the hidden lines result from a square hole through the middle of the object.



The hidden lines in this example are there because a part of one corner of the front surface was cut away, or “recessed.”

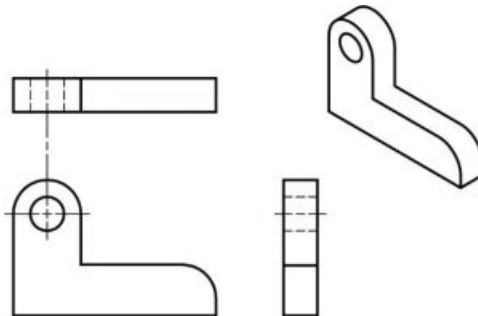
Hidden Surfaces

Directions: Draw the hidden lines which are missing in the views below. Each problem has one incomplete view.

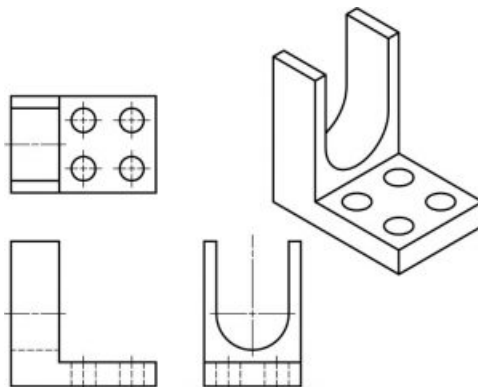


Curved Surfaces

Curved surfaces are perhaps tricky to “see” until you remember that the curve is only shown in one view. You must put the curve in the other views yourself, through visualization. Try to think that when there is a sharp change of direction like at a corner, then that will produce a line visible in another view. When the change of direction is smooth like a curve, no line will be seen.

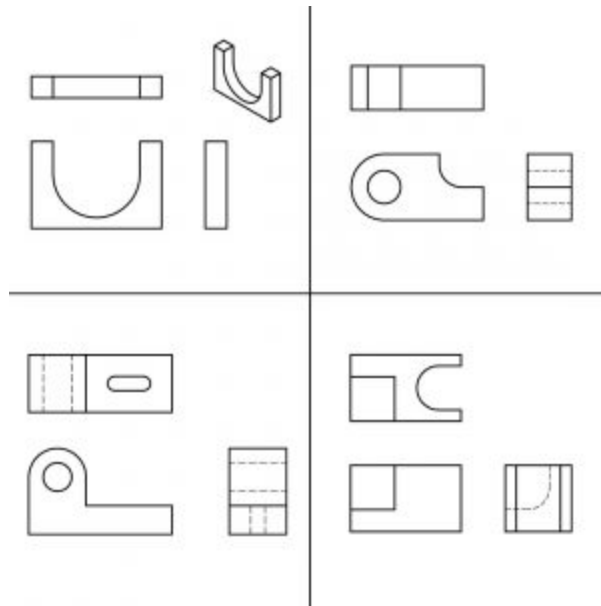


Here's another example of curved surfaces:



Curved surfaces exercise.

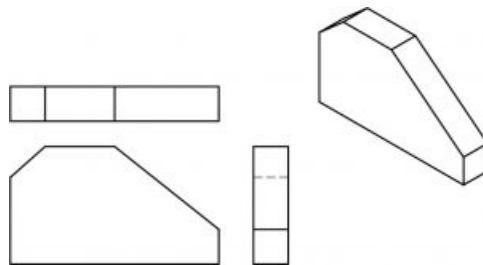
Directions: Draw the lines which are missing in the views below. Each problem has one incomplete view. Do not draw center lines.



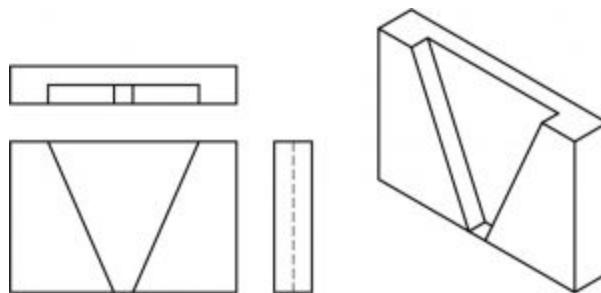
Inclined Surfaces

Inclined surfaces are those which are at an angle, or slanted. In other words, they are surfaces which are neither horizontal nor vertical. In viewing orthographic drawings you need to be alert to angles and inclined surfaces, for they are often found on the prints you will be reading later.

Notice the hidden line in the right view created by the inclined surface on this object:

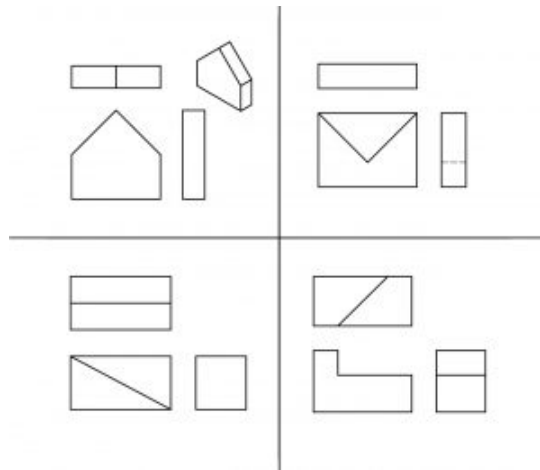


Here is an object with two inclined surfaces.



Inclined surfaces exercise

Directions: Draw the lines which are missing in the views below. Each problem has one incomplete view.



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2: Auxiliary Views

When an object has a slanted or inclined surface, it usually is not possible to show the inclined surface in an orthographic drawing without distortion. To present a more accurate description of any inclined surface, an additional view, known as an auxiliary view, is usually required.

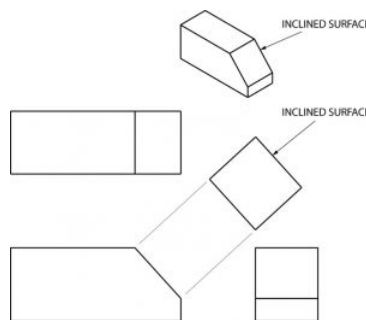
An auxiliary view is simply a “helper” view, which shows the slanted part of the object as it actually is. It turns, or projects, the object so that the true size and shape of the surface (or surfaces) are seen as they actually are.

Auxiliary views are commonly found on many types of industrial drawings.

Front View Auxiliaries

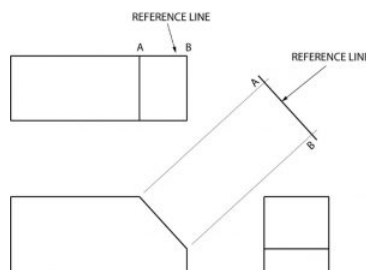
There are three basic type of auxiliary views. In the first type, the auxiliary view is projected from the front view of a three view (orthographic) drawing. In the second and third types of drawings, the auxiliary views are projected from the top and side views.

Here is a front view auxiliary of a simple object with an inclined surface.

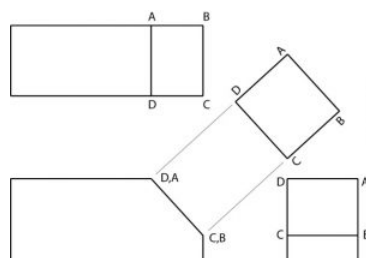


Notice that the projection lines are perpendicular to the slanted surface of the first view, and that only the slanted surface of the object is shown in the auxiliary view. The rest of the object is omitted, however, for clarification portions of the adjacent surfaces are sometimes shown. Also, notice that the slanted surfaces of the top and side views are shortened because of distortion, whereas the surface of the auxiliary view is true, or actual size.

To sketch an auxiliary view, you begin with orthographic views of the object and add projection lines perpendicular (90°) to the slanted surface, adding a reference line any convenient distance from the view with the slanted surface.



Next, the distance CB on the auxiliary view is made the same length as the related distance in one of the orthographic views; in this example it's the side view. This completes the auxiliary view.

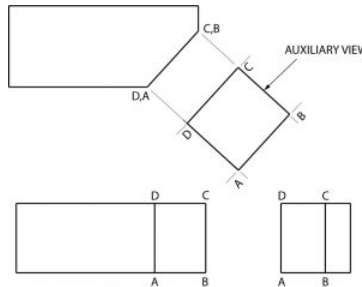


Top View Auxiliaries

A top view auxiliary is developed in the same way as a front view auxiliary, except that the auxiliary is projected from the top view.

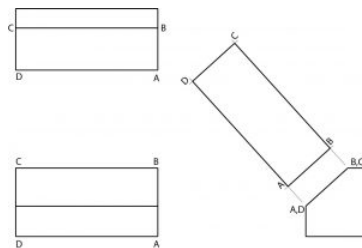
Whether the auxiliary view is to be projected from the front, top, or side view depends on the position of the object, or which surface of the object is slanted. In this example, the top view is slanted. Therefore the auxiliary view must be projected from the top view.

Again, notice how the angled surfaces shown in the front and side views are not shown in true length.



Side View Auxiliary

Side view auxiliaries are drawn in the same way as front and top view auxiliaries. Again, where the auxiliary view is to be projected depends upon the position of the object or which surface of the object is slanted.



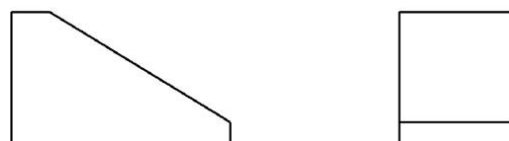
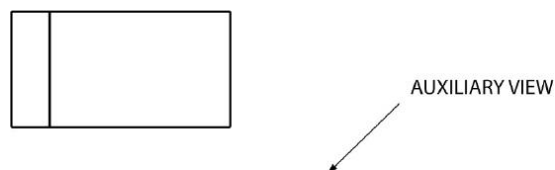
Obviously, these are very basic auxiliary view examples and are presented to introduce you to the concept of auxiliary views.

As objects with inclined surfaces become more complex, auxiliary views provide a means of presenting objects in their true size and shape.

Sketching Auxiliary Views

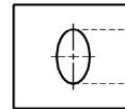
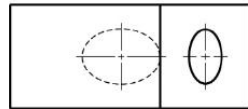
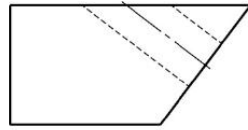
The following problems require an auxiliary view to be complete. Sketch the auxiliary views required in the spaces provided.

Drawing practice 1



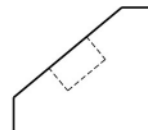
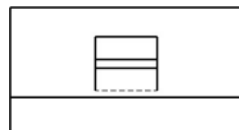
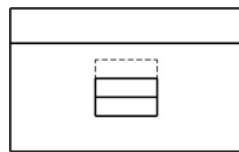
Drawing practice 2

In this problem, a round hole is centered on the slanted surface and drilled through the object. The hole appears elliptical in the front and side views because of distortion. It will appear in its true shape on the auxiliary view. Remember that the auxiliary is developed from the view with the slanted surface. Complete the auxiliary view.



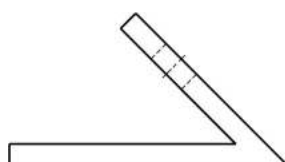
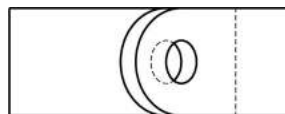
Drawing practice 3

In this problem, a square hole has been cut part way into the object. Complete the auxiliary view.



Quiz

Directions: Complete the auxiliary view in the space provided.



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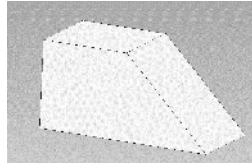
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3: Machined Features

The machined features in this section are common terms related to basic industry processes. These terms are often found on prints. For a better understanding of these processes, look at the models of machined features in the Print Reading Lab.

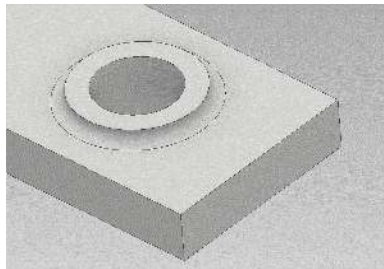
Bevel

A surface cut at an angle. In regard to welding, a bevel will normally end up being a surface prep for a weld.



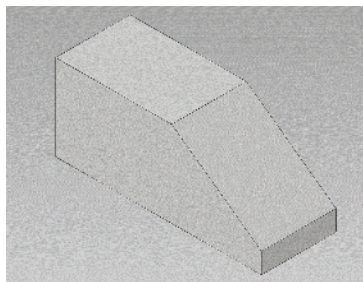
Boss

A circular pad on forgings or castings, which project out from that body of the part. The surface of the boss is machined smooth for a bolt head to seat on and it has a hole drilled through to accommodate the bolt shank.



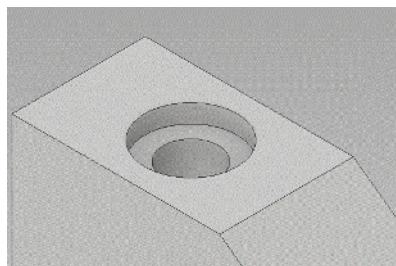
Chamfer

A process of cutting away a sharp external corner or edge. Not for welding.



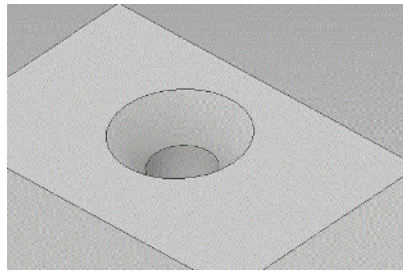
Counterbore

To enlarge drilled hole to a given diameter and depth. Usually done for recessing a bolt head.



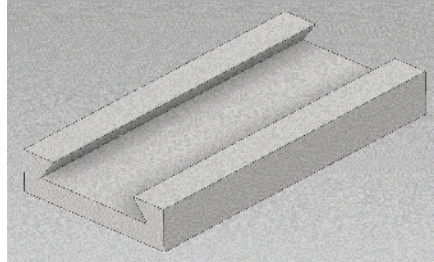
Countersink

To machine a conical depression in a drilled hole for recessing flathead screws or bolts.



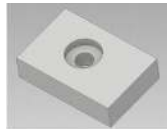
Dovetail

A slot of any depth and width, which has angled sides.



Quiz

Directions: Name the machined features shown below.



1. _____



2. _____



3. _____



4. _____



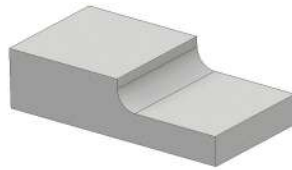
5. _____



6. _____

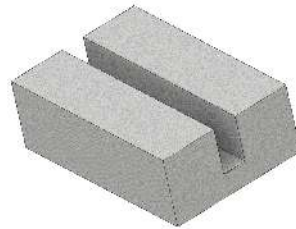
Fillet

A small radius filling formed between the inside angle of two surfaces.



Kerf

The narrow slot formed by removing material while sawing or other machining.



Keyway

A narrow groove or slot cut in the shaft hole of a sleeve or hub for accommodating a key.

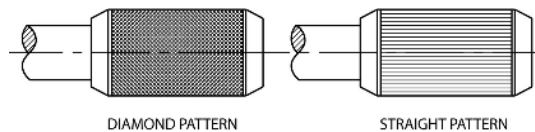
Keyseat

A narrow groove or slot cut in a shaft for accommodating a key.



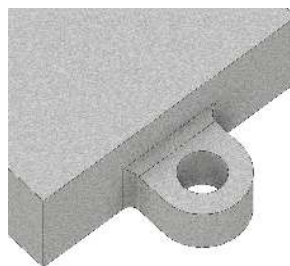
Knurl

To uniformly roughen with a diamond or straight pattern a cylindrical or flat surface.



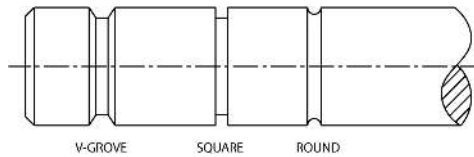
Lug

A piece projecting out from the body of a part. Usually rectangular in cross section with a hole or slot in it.



Neck

To machine a narrow groove on a cylindrical part or object.

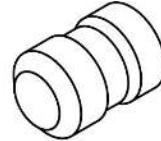


Quiz

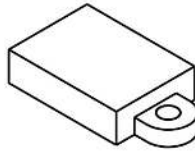
Directions: Name the machined features shown below. Check your answer.



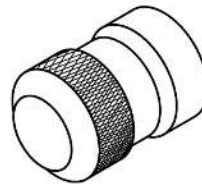
1. _____



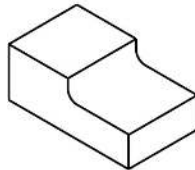
4. _____



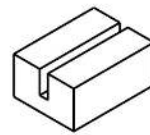
2. _____



5. _____



3. _____



6. _____

Additional features and shapes.

Pad

A slightly raised surface projecting out from the body of a part. The pad surface can be of any size or shape. (Remember, bosses can only be round)



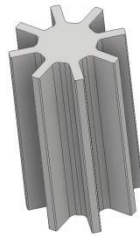
Round

A small radius rounded outside corner formed between two surfaces.



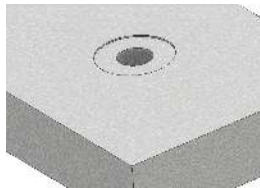
Spline

A gear-like serrated surface on a shaft. Take the place of a key when more torque strength is required.



Spotface

A round surface on a casting or forging for a bolt head. Usually about 1/16" deep.



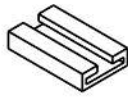
T-Slot

A slot of any dimensions to resemble a "T".

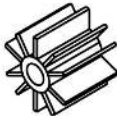


Quiz

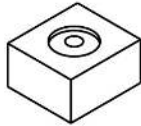
Directions: Name that machined features shown below. Check your Answers.



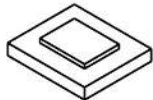
1. _____



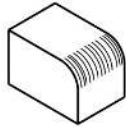
2. _____



3. _____



4. _____



5. _____

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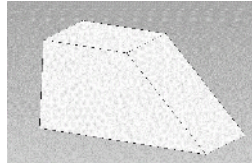
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4: Machined Features

The machined features in this section are common terms related to basic industry processes. These terms are often found on prints. For a better understanding of these processes, look at the models of machined features in the Print Reading Lab.

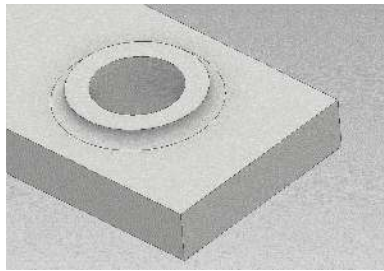
Bevel

A surface cut at an angle. In regard to welding, a bevel will normally end up being a surface prep for a weld.



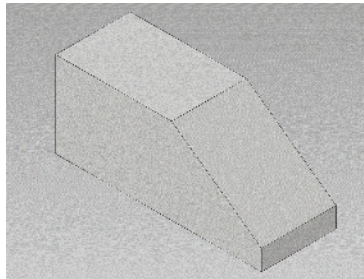
Boss

A circular pad on forgings or castings, which project out from that body of the part. The surface of the boss is machined smooth for a bolt head to seat on and it has a hole drilled through to accommodate the bolt shank.



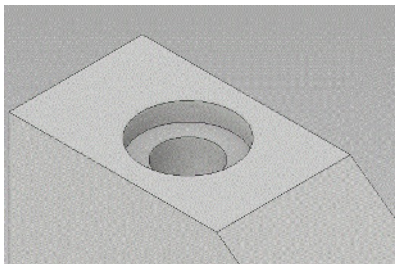
Chamfer

A process of cutting away a sharp external corner or edge. Not for welding.



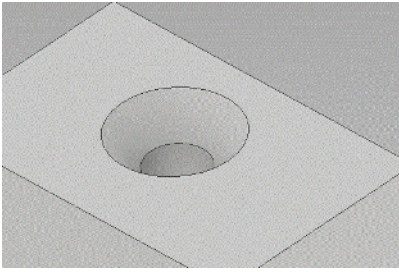
Counterbore

To enlarge drilled hole to a given diameter and depth. Usually done for recessing a bolt head.



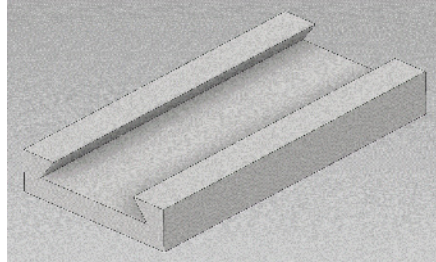
Countersink

To machine a conical depression in a drilled hole for recessing flathead screws or bolts.



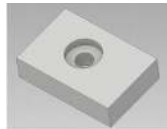
Dovetail

A slot of any depth and width, which has angled sides.



Quiz

Directions: Name the machined features shown below.



1. _____



2. _____



3. _____



4. _____



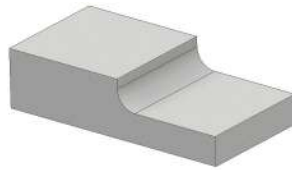
5. _____



6. _____

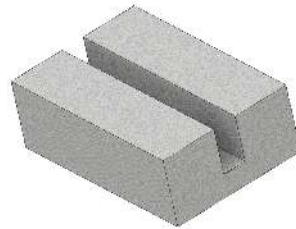
Fillet

A small radius filling formed between the inside angle of two surfaces.



Kerf

The narrow slot formed by removing material while sawing or other machining.



Keyway

A narrow groove or slot cut in the shaft hole of a sleeve or hub for accommodating a key.

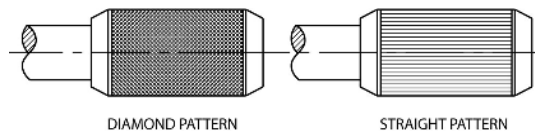
Keyseat

A narrow groove or slot cut in a shaft for accommodating a key.



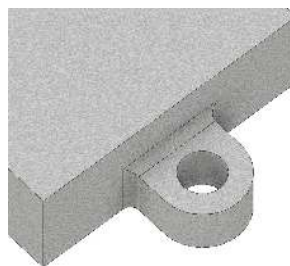
Knurl

To uniformly roughen with a diamond or straight pattern a cylindrical or flat surface.



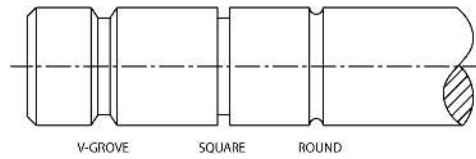
Lug

A piece projecting out from the body of a part. Usually rectangular in cross section with a hole or slot in it.



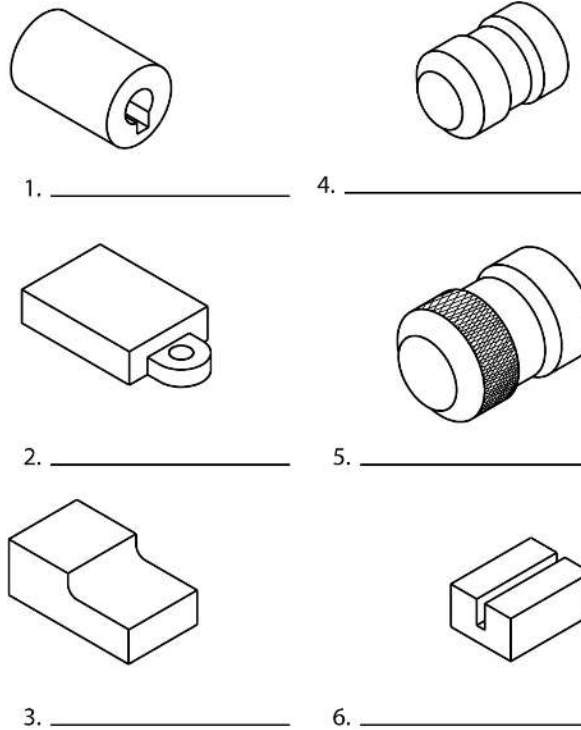
Neck

To machine a narrow groove on a cylindrical part or object.



Quiz

Directions: Name the machined features shown below. Check your answer.



Additional features and shapes.

Pad

A slightly raised surface projecting out from the body of a part. The pad surface can be of any size or shape. (Remember, bosses can only be round)



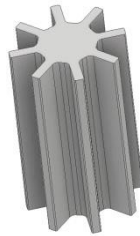
Round

A small radius rounded outside corner formed between two surfaces.



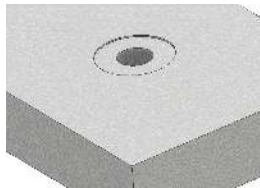
Spline

A gear-like serrated surface on a shaft. Take the place of a key when more torque strength is required.



Spotface

A round surface on a casting or forging for a bolt head. Usually about 1/16" deep.



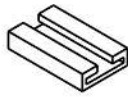
T-Slot

A slot of any dimensions to resemble a "T".

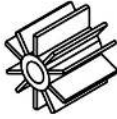


Quiz

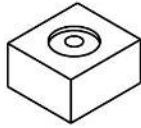
Directions: Name that machined features shown below. Check your Answers.



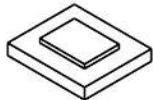
1. _____



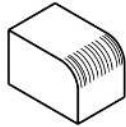
2. _____



3. _____



4. _____



5. _____

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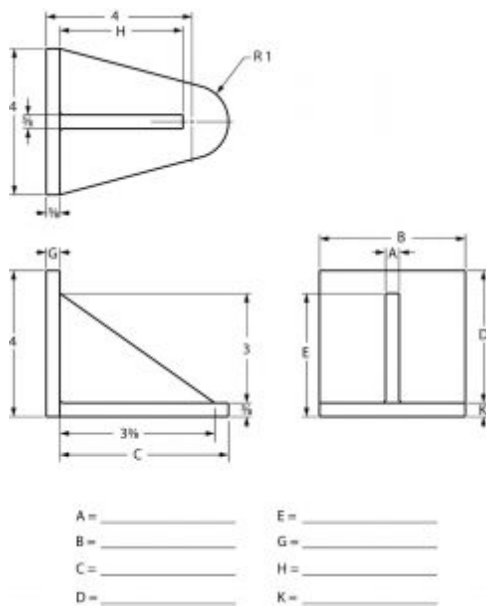
5: Print Interpretation

This final section introduces basic print reading. Because machine drawings are used to some extent in nearly every trade, the working drawings used in this section are all machine drawings.

The purpose of this package is to provide an opportunity to put your fundamental knowledge of print reading to use before you go on to more specialized and advanced print reading activities.

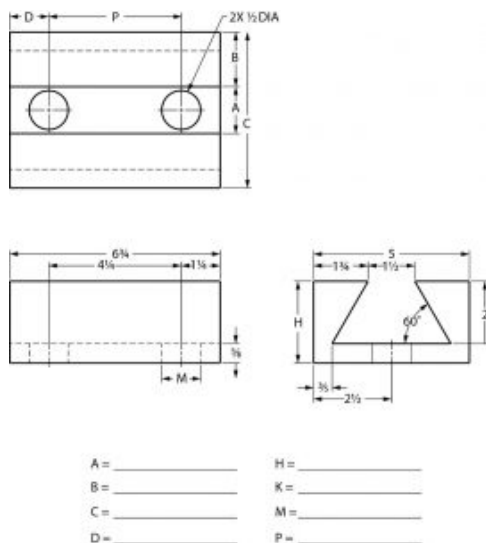
Exercise 1

Study the print below and fill in the related dimensions.



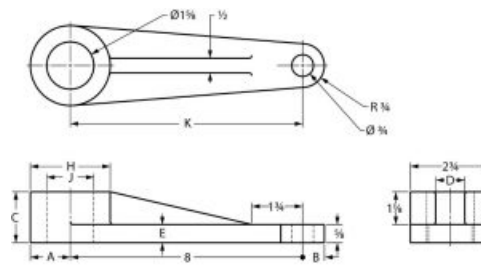
Exercise 2

Study the print below and fill in the related dimensions.



Exercise 3

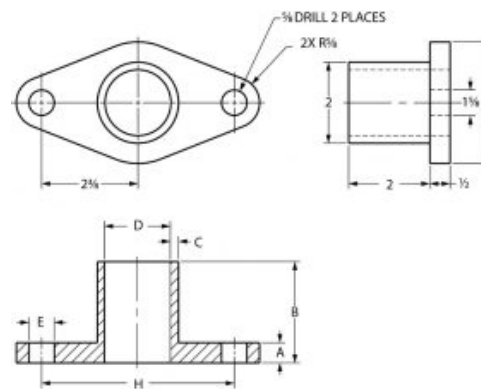
Study the print below and fill in the related dimensions.



A = _____ E = _____
B = _____ H = _____
C = _____ J = _____
D = _____ K = _____

Exercise 4

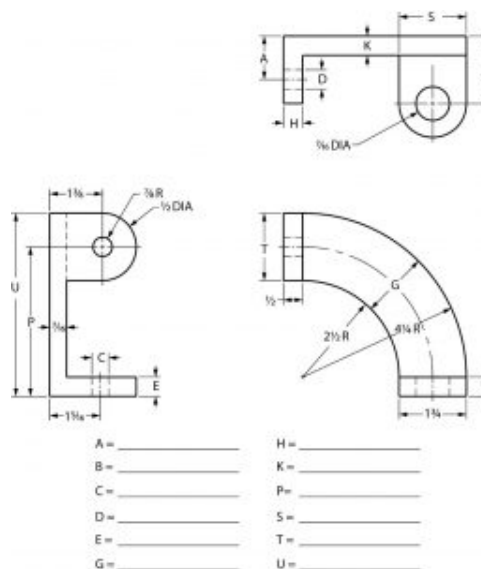
Study the print below and fill in the related dimensions.



A = _____ D = _____
B = _____ E = _____
C = _____ H = _____

Exercise 5

Study the print below and fill in the related dimensions.



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6: Measuring, Marking, Leveling and Layout Tools

Measuring and marking tools are common to multiple trades and ensure accuracy and quality craftsmanship in the building and construction process. Measuring devices are available in fractional and metric, with classic western construction practices adopting the fractional inch configuration. While some measuring tools have the actual fractions printed ($1/2"$, $1/4"$, and $1/8"$) next to each corresponding mark on the scale, many do not and the ability to read the scale without the printed fractions may take time to develop. This is one of many reasons for the adage: “Measure twice, cut once”, which is a good habit to develop and will help to avoid costly mistakes such as over cutting and wasting material or under cutting, resulting in having to remeasure and cut the material again.

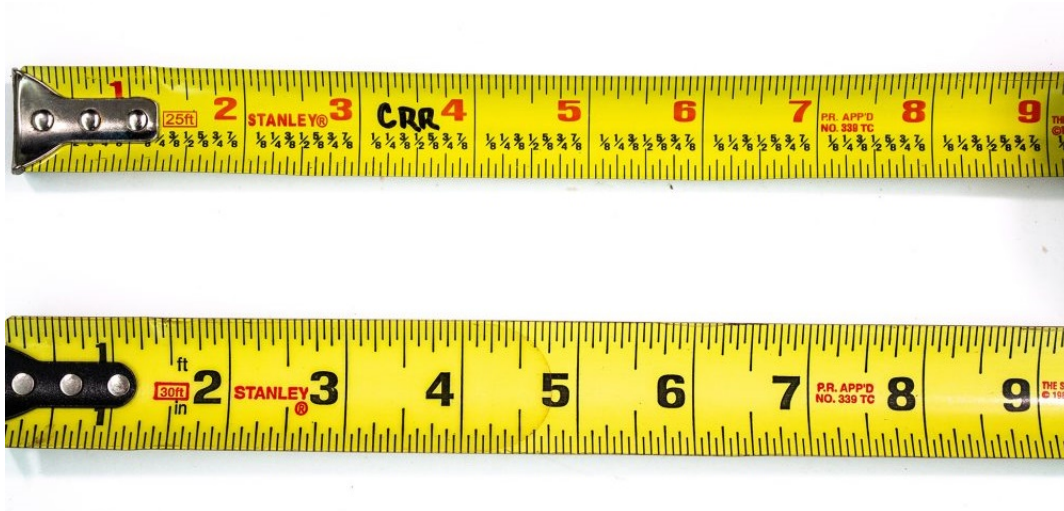


Figure 6.1: Fractional Marked & Unmarked Tape Measures by Gwen Arkin is licensed under CC BY 4.0

While many of the following tools may be considered “carpentry” tools, the majority are regularly used in most all trades. Apparent common items to all trades include measuring devices like tape measures, rulers, and basic estimation tools; plumbers and electricians will use spirit levels, builders levels, and angle finders to ensure that piping is at an appropriate grade or conduit, fixtures, and other items are installed to meet industry codes.

Levels are used to check for **level (horizontal)** and **plumb (vertical)**. Trades persons will often use a combination of various levels and squares to complete a project.

Squares are used for layout work to mark square (90° /right angle) and other angles commonly used in building and construction trades indicated on specific types of squares. They are also used to check for squareness and other angles during assembly. Squares can be made of inexpensive molded plastic, lightweight aluminum or durable steel.

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7: The Language of Lines

You have heard the saying, “A picture is worth a thousand words”. This statement is particularly true in regards to technical drawings.

It would be almost impossible for an engineer, designer, or architect to describe in words the shape, size, and relationship of a complex object. Therefore, drawings have become the universal language used by engineers, designers, technicians, as well as craftsmen, to communicate the Information necessary to build, assemble and service the products of industry.

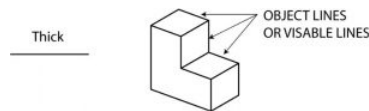
It is Important to remember, as you study Print Reading, that you are learning to communicate with the graphic language used by Industry: Lines are part of that language.

Since technical drawings are made of lines, it is logical that the first step in learning to “read” a drawing is to learn the meaning of each kind of line. Generally, there are 11 basic types of lines. Each kind of line has a definite form and “weight”. Weight refers to line thickness or width. When combined in a drawing, lines provide part of the Information needed to understand the print.

Being able to interpret a blueprint and accurately build objects is a needed skill to become successful in all trade crafts. It is a skill, like many others you will learn, and it will take time and practice to fully understand and become proficient.

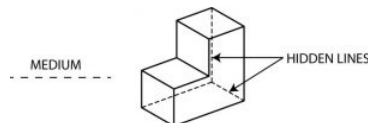
Object Line

A **visible line**, or **object line** is a thick continuous line, used to outline the visible edges or contours of an object.



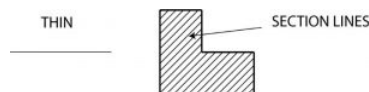
Hidden Line

A **hidden line**, also known as a **hidden object line** is a medium weight line, made of short dashes about 1/8” long with 1/16” gaps, to show edges, surfaces and corners which cannot be seen. Sometimes they are used to make a drawing easier to understand. Often they are omitted in an isometric view.



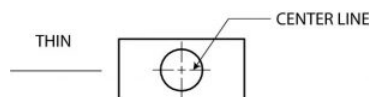
Section Line

Section lines are used to show the cut surfaces of an object in section views. They are fine, dark lines. Various types of section lines may indicate the type of material cut by the cutting plane line.



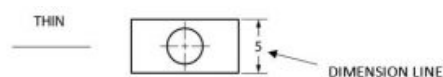
Center Line

Center lines are used to indicate the centers of holes, arcs, and symmetrical objects. They are very thin (size), long-short-long kinds of lines.



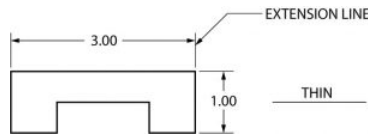
Dimension Line

Dimension lines are thin and are used to show the actual size of an object. There are arrowheads at both end that terminate at the extension lines.



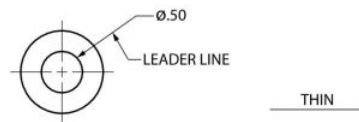
Extension Line

Extension lines are also thin lines, showing the limits of dimensions. Dimension line arrowheads touch extension lines.



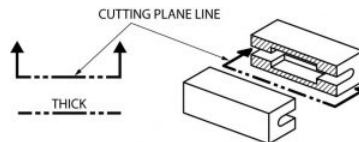
Leader Line

Leaders are more thin lines used to point to an area of a drawing requiring a note for explanation. They are preferably drawn at a 45° angles.



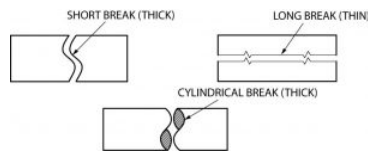
Cutting Plane Line

A cutting plane line (very heavy) helps to show the internal shape at a part or assembly by slicing through the object.



Break Line

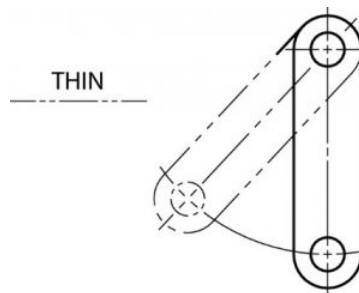
There are three kinds of break lines used in drawings. They are used to remove, or 'break out' part of a drawing for clarity, and also to shorten objects which have the same shape throughout their length and may be too long to place on the drawing.



Short and long break lines are used for flat surfaces. Cylindrical are used on rods, dowels, etc.

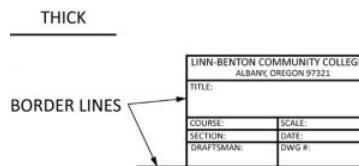
Phantom Line

Phantom lines are long-short-short-long lines most often used to show the travel or movement of an object or a part in alternate positions. It can also be used to show adjacent objects or features.



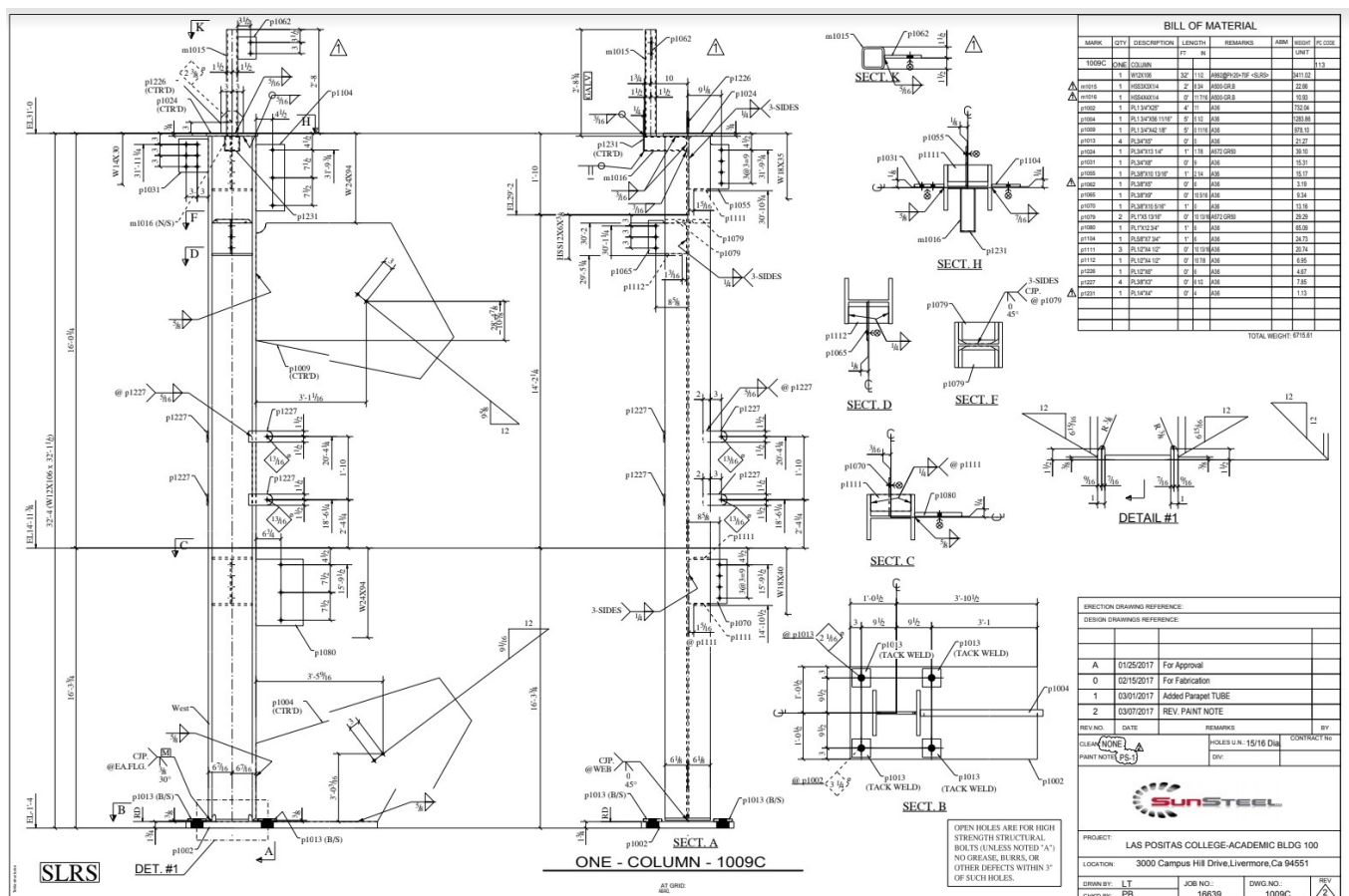
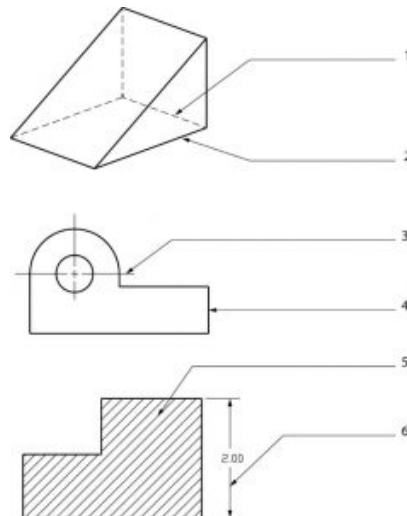
Border Line

Borderlines are very thick, continuous lines used to show the boundary of the drawing or to separate different objects drawn on one sheet. They are also used to separate the title block from the rest of the drawing.



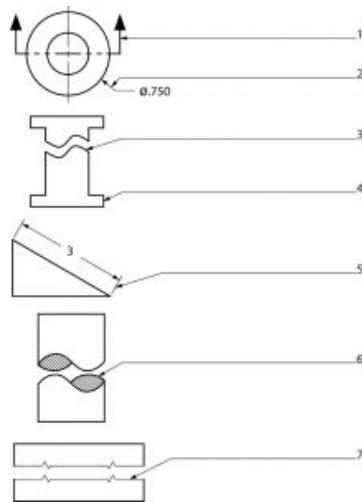
Quiz...

Directions: Name the types of lines shown below. Check your own answers

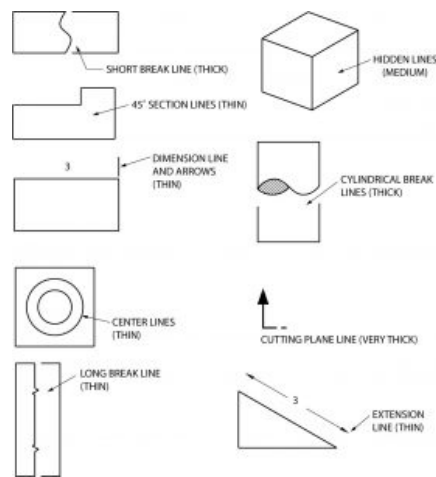


Identify the various line types used in this drawing. (instructor will provide a copy of this drawing)

Name the types of lines shown below. Check your own answers.



Directions: Draw and identify the lines needed to complete the figures as indicated.



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8: Scaling

Whether or not you need to review these fundamentals, there is one important thing to remember about getting measurements from a print. If you need a dimension that is unclear or is not given, do not measure the print! Since prints shrink, stretch, and may not be drawn to scale, you can easily come up with some very inaccurate dimensions.

Scale Measurement

A drawing of an object may be the same size as the object (full size), or it may be larger or smaller than the object. In most cases, if it is not drawn full size, the drawing is made smaller than the object. This is done primarily for the convenience of the users of the drawings. After all, who wants to carry around a full size drawing of a locomotive? Obviously, with an object as small as a wristwatch, it would be necessary to draw to a larger scale.

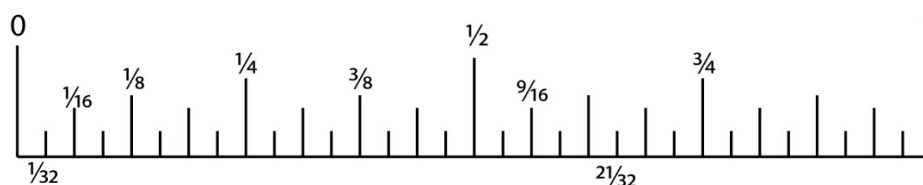
A machine part, for example, may be half the size ($1/2"=1"$); a building may be drawn $1/48$ size ($1/4"=1'-0"$); a map may be drawn $1/200$ size ($1"=100'-0"$); and a gear in that wristwatch may be ten-times size ($10"=1"$).

There are numerous scales for different needs. Since each occupational group has their own frequently used scales, some practice or basics review will help you to work with the scales used in your technology.

Full Scale

Full scale is simply letting one inch on a ruler, steel rule, or draftsman's scale equal one inch on the actual object. Rules of this kind are usually divided into $1/16"$ or $1/32"$ units. The first measurement exercise will be with full size. If you can measure accurately in full scale, you may want to skip ahead.

Here is a "big Inch". Each space equals $1/32"$. If you have not worked with accurate measurement, spend some time studying it.

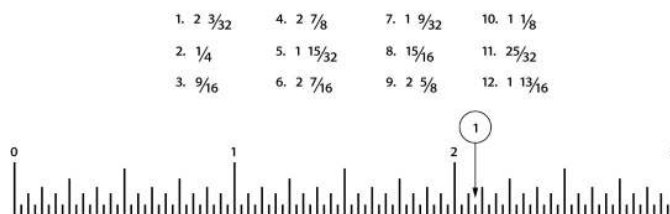


Measurement practice: on the scale above, locate the following fractions:

- | | |
|------------|------------|
| 1. $5/8$ | 1. $7/8$ |
| 2. $3/16$ | 2. $11/16$ |
| 3. $7/32$ | 3. $5/16$ |
| 4. $15/16$ | 4. $31/32$ |
| 5. $25/32$ | 5. $19/32$ |

Directions:

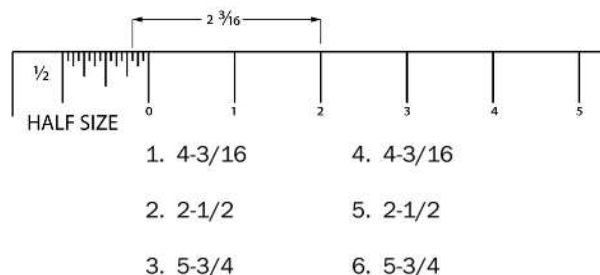
Each of the fractions below is numbered. Write that number above the scale and point with an arrow where the fraction is located. Number 1 has been completed.



Half Size

The principle of half size measurements on a drawing is simply letting a unit, such as $\frac{1}{2}$ " on the scale, represent a larger unit such as 1" on the drawing. If the drawing is properly labeled, the words HALF SIZE or $\frac{1}{2}" = 1'$ will appear in the title block.

Using the half-size scale is not difficult, but it does take some practice. To measure a distance of $2\frac{3}{16}$ " you look first for the 2, then go backwards to the zero and count off another $\frac{3}{16}$ ". You measure this way for each dimension that has a fraction. Whole numbers (numbers without fractions) are measured in the usual way.



Next, locate a half size scale (available in the lab) and measure the lines below to the nearest $\frac{1}{32}$ of an inch. Write the length of the line in the space provided.

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____

Because paper is dimensionally unstable due to humidity, exact answers to this half size measurement practice cannot be given. That is another reminder that it's poor practice to measure from a piece of paper.

Quarter Size

Quarter size is used and read in a similar way to half size except that each unit, such as a quarter of an inch, represents a larger unit, such as one inch. If the drawing is properly labeled the words QUARTER SIZE, QUARTER.SCALE, or $\frac{1}{4}" = 1'$ will appear in the title block.

The quarter size scale is used in a similar manner as the half size scale.

For quarter size practice, draw lines in the area provided to the required length. Have another student or the lab instructor check your lines for accuracy. (A $\frac{1}{4}" = 1'$ scale is available in the lab.)

1. 1"
2. 4"
3. 7 1/4"
4. 16 3/8"
5. 5 3/16"
6. 13"
7. 10 7/8"
8. 2 5/16"

Quiz

For this quiz, you will be given an object to measure with a ruler or tape measure. You will record measurements in full scale and then draw each length in $\frac{1}{2}$ and $\frac{1}{4}$ scale.

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Sample Word 1 | Sample Definition 1

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